

Complements of Operational Research

4.1 Replacement theory: equipment that wears out

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When should a piece of equipment be replaced?

Two types of problem:

1. **Gradual wear** — the equipment keeps working, but becomes more and more expensive: maintenance, fuel, breakdowns, loss of value. Vehicles, machines, computers. (4.1 and 4.2)
2. **Sudden failure** — it works until it fails for good. Lamps, valves, pumps, sensors. (4.3)

The trade-off (gradual wear)

Keeping it longer spreads the purchase cost over more years, but the running costs go up. Somewhere there is an **optimal point**.

Where it is used

- fleets (vans, buses, forklifts);
- industrial and hospital equipment;
- IT equipment (laptop renewal);
- infrastructure (pumps, valves, lighting).

Question: after how many years should the equipment be replaced by an identical one, so that the cost *per year* is minimum?

The example of this chapter: the delivery van

A delivery company buys **diesel** vans for $P = 30\,000$ €. The running costs (maintenance, breakdowns, fuel above normal) and the resale value at the end of each year are in the table.

If it sells after n years, the van has cost $P - S_n$ (loss of value) plus $C_1 + \dots + C_n$.

Assumptions (4.1)

- it is replaced by an identical van, and so on;
- the value of money does not change over time (see 4.2);
- costs are known and paid in the year they occur.

Year n	1	2	3	4	5	6	7	8
Running cost C_n (€)	3000	3600	4400	5400	6400	7600	9800	12500
Resale at end of year, S_n (€)	22000	17000	13500	11000	9000	7500	6200	5000

Question: after how many years should the van be sold and a new one bought?

Average annual cost

Criterion

$$A(n) = \frac{P - S_n + \sum_{i=1}^n C_i}{n}$$

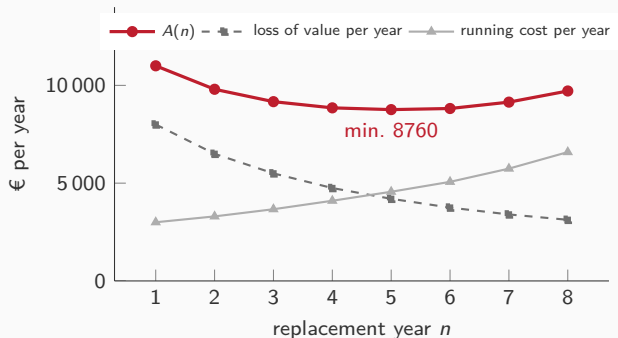
Replace after n^* years, with $A(n^*)$ minimum.

If there is only a scrap value S (the same in every year), use $P - S$.

n	C_n	$\sum C_i$	$P - S_n$	Total	$A(n)$
1	3000	3000	8000	11000	11000.00
2	3600	6600	13000	19600	9800.00
3	4400	11000	16500	27500	9166.67
4	5400	16400	19000	35400	8850.00
5	6400	22800	21000	43800	8760.00
6	7600	30400	22500	52900	8816.67
7	9800	40200	23800	64000	9142.86
8	12500	52700	25000	77700	9712.50

Replace the van after 5 years: average cost of 8760 € per year.

Why there is a minimum



$A(n)$ is the sum of two terms:

- **loss of value per year**, $(P - S_n)/n$: **decreases** — the purchase is spread out;
- **running cost per year**, $\sum C_i/n$: **increases** — the van gets older.

The bottom is **flat**: between 4 and 6 years the difference is about 1%. In practice, the decision is robust to small errors in the data — and other criteria (image, emissions) can break the tie.

The rule: replace when next year is above the average

Rule

Keeping it one more year costs

$$M_{n+1} = C_{n+1} + (S_n - S_{n+1})$$

(running cost + the drop in resale value). Replace at the end of year n when

$$M_{n+1} > A(n).$$

With no resale (scrap only), $M_{n+1} = C_{n+1}$: replace when the cost of next year exceeds the average so far.

For the van

- Year 5: $M_5 = 6400 + 2000 = 8400 < A(4) = 8850 \rightarrow$ it pays to keep it for the 5th year (it lowers the average).
- Year 6: $M_6 = 7600 + 1500 = 9100 > A(5) = 8760 \rightarrow$ the 6th year would raise the average: **replace at the end of year 5.**

Intuition: if an extra year costs more than the average, it pulls the average up.

Two machines: the electric van arrives

A supplier offers an **electric** van: $P = 44\,000$ €, more expensive, but cheaper to run.

Year n	1	2	3	4	5	6	7	8
C_n	1600	1900	2400	3100	4000	5400	7000	9000
S_n	32000	25500	21000	17500	14800	12500	10600	9000
$A(n)$	13600	11000	9633	8875	8440	8317	8400	8675

The electric van, replaced after 6 years, costs **8316.67 €/year** — less than the diesel one at its optimum (8760).

Question: the company has a diesel van that is **3 years** old. Should it switch to the electric one now? In how long?

Rule

Keep the old machine while the cost of one more year of it is **lower** than the minimum average cost of the new one.

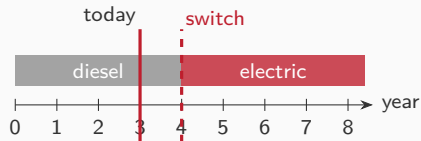
When to switch

Cost of keeping the diesel van one more year,

$$M_{n+1} = C_{n+1} + (S_n - S_{n+1}):$$

Diesel van's year	M	Decision
4th	$5400 + 2500 = 7900$	< 8316.67 : keep
5th	$6400 + 2000 = 8400$	> 8316.67 : switch

Keep the diesel van one more year and switch at the end of the 4th year.



Note: on its own, the diesel van would be replaced after 5 years. With a cheaper alternative, it is switched one year earlier. The decision compares the marginal cost of the old one with the average cost of the new one — not the two averages.

```
def custo_medio(P, C, S):  
    """P: purchase; C, S: lists per year (running cost, resale)."""  
    A, soma = [], 0  
    for n, (c, s) in enumerate(zip(C, S), start=1):  
        soma += c  
        A.append((P - s + soma) / n)  
    return A  
  
P = 30000  
C = [3000, 3600, 4400, 5400, 6400, 7600, 9800, 12500]  
S = [22000, 17000, 13500, 11000, 9000, 7500, 6200, 5000]  
A = custo_medio(P, C, S)  
n = A.index(min(A)) + 1  
print(n, round(A[n-1], 2))      # 5 8760.0
```

A spreadsheet does the same: one column per quantity and the MIN function.

Full file (average cost and marginal rule):
code/python/04_replacement/04_1_wear/
desgaste.py (function names in Portuguese,
as in the course code).

To solve in class

Example 1

Machine with purchase cost 7100 and scrap value 100.

Year	1	2	3	4	5	6	7	8
C_n	200	350	500	700	1000	1300	1700	2100

Example 2

Machine with purchase cost 24400 and scrap value 400.

Year	1	2	3	4	5	6	7	8
C_n	400	1000	1600	2400	3600	5000	6400	8000

When should it be replaced?

Example 3 — two machines

Machine	A	B
Purchase cost	45000	55000
Running cost in the 1st year	2000	3000
Increase in the following years	10000	4000
Resale value	0	0

- a) Machine A is in operation. Should it be replaced by B? When?
- b) If one were to be bought now, which of the two should be purchased?

KEY TAKEAWAYS — 4.1

1. Average annual cost: $A(n) = (P - S_n + \sum_{i \leq n} C_i) / n$; replace after n^* with A minimum.
2. Rule: replace when the cost of one more year, $C_{n+1} + (S_n - S_{n+1})$, exceeds $A(n)$.
3. Two machines: keep the old one while the cost of one more year of it is lower than the **minimum** average cost of the new one.
4. The minimum is usually flat: the decision is robust; always check the neighbouring years.

Next: 4.2 What if money has a time value? (discount rate)

References and resources

References Full list: *Course bibliography*.

- Kasana, H. S., Kumar, K. D., *Introductory Operations Research*, Springer, 2004 — ch. 11, Replacement Theory.
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017 — equipment replacement models.
- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021.
- Jardine, A. K. S., Tsang, A. H. C., *Maintenance, Replacement, and Reliability*, 2nd ed., CRC Press, 2013.

Resources

- Exercise sheet 4. `code/python/04_replacement/04_1_wear/desgaste.py`.